CELLSCOPE DRIVE-TEST & RF COVERAGE REPORT

Automated Native Android Cellular Field Assessment • Lead: Sarkar Tuhassinul Arnob

1. Session Executive Summary

Session Name:	Survey_20261018_101517	**Start Timestamp:**	2026-10-18 10:15:17 (BST)
Duration:	00:00:02	**Mobility Mode:**	DYNAMIC DRIVE
Route Distance:	23.28 km	**Max Speed:**	66.5 km/h
Sample Count:	2131 measurements	**Mean RSRP:**	-80.3 dBm
GPS Availability:	98.8% (2105 valid)	**Radio Availability:**	100.0%
Coverage Adequacy:	94.6%	**Serving Cells Observed:**	41 distinct PCIs

2. RF Signal Quality Distribution (3GPP TS 36.133)



Excellent (≥ -80 dBm) **38.2% (814)**

Good (-90 to -81 dBm) **32.8% (698)**

Fair (-100 to -91 dBm) **18.4% (393)**

Poor (< -100 dBm) **10.6% (226)**

3. Handover & Serving-Cell Dwell Analytics

Total Cell Transitions: 65 | Confirmed Handovers: 26 | Reselections: 1

Transition Density: 2.79 transitions/km | Signal Loss Reselections: 0

PCI	Samples	Dwell Time	Session Share	Mean RSRP
PCI 492	154	2m 42s	7.2%	-80.2 dBm
PCI 144	133	2m 20s	6.3%	-77.8 dBm
PCI 1	132	2m 19s	6.2%	-86.0 dBm
PCI 427	129	2m 16s	6.1%	-66.0 dBm
PCI 438	121	2m 07s	5.7%	-82.0 dBm
PCI 331	94	1m 39s	4.4%	-44.0 dBm
PCI 111	94	1m 36s	4.3%	-77.0 dBm
PCI 458	87	1m 30s	4.1%	-90.0 dBm
PCI 223	80	1m 23s	3.7%	-44.0 dBm
PCI 184	75	1m 19s	3.5%	-82.2 dBm
PCI 362	71	1m 14s	3.3%	-88.5 dBm
PCI 122	69	1m 13s	3.3%	-101.2 dBm
PCI 25	68	1m 10s	3.2%	-99.5 dBm
PCI 361	67	1m 10s	3.1%	-89.1 dBm
PCI 267	67	1m 10s	3.1%	-87.0 dBm
PCI 13	67	1m 09s	3.1%	-86.6 dBm
PCI 454	60	1m 03s	2.8%	-68.0 dBm
PCI 15	58	1m 01s	2.7%	-93.0 dBm
PCI 50	57	1m 00s	2.7%	-80.2 dBm
PCI 360	55	0m 58s	2.6%	-86.3 dBm
PCI 37	47	0m 49s	2.2%	-65.7 dBm
PCI 4	46	0m 47s	2.1%	-87.0 dBm
PCI 493	45	0m 47s	2.1%	-90.5 dBm
PCI 219	42	0m 43s	2.0%	-110.0 dBm
PCI 26	34	0m 35s	1.6%	-89.0 dBm
PCI 67	32	0m 34s	1.5%	-97.0 dBm
PCI 58	24	0m 25s	1.1%	-91.1 dBm
PCI 40	20	0m 20s	0.9%	-109.0 dBm
PCI 42	17	0m 18s	0.8%	-44.0 dBm
PCI 51	17	0m 17s	0.8%	-44.0 dBm
PCI 274	13	0m 13s	0.6%	-108.0 dBm
PCI 113	10	0m 10s	0.5%	-111.0 dBm
PCI 183	10	0m 10s	0.5%	-86.0 dBm
PCI 456	7	0m 07s	0.3%	-44.0 dBm
PCI 29	6	0m 06s	0.3%	-44.0 dBm

PCI	Samples	Dwell Time	Session Share	Mean RSRP
PCI 18	5	0m 05s	0.2%	-44.0 dBm
PCI 24	5	0m 05s	0.2%	-105.0 dBm
PCI 9	4	0m 04s	0.2%	-44.0 dBm
PCI 299	3	0m 03s	0.1%	-98.0 dBm
PCI 147	3	0m 03s	0.1%	-44.0 dBm
PCI 153	3	0m 03s	0.1%	-44.0 dBm

4. Identified Weak-Coverage Zones & Remediation Feasibility

Zone #1: Length ~34m | Mean RSRP: -105.0 dBm | Min: -105 dBm
Diagnosis: Moderate Shadowing & Cell Edge: Signal fading near serving boundary (PCI 4).
Recommended Action: [ANTENNA_OPTIMIZATION] Sector Power Boost / Tilt Refinement (Feasibility: 75%)
Rationale: Adjusting RS (Reference Signal) power allocation or mechanical tilt may extend effective coverage radius into this edge zone.

Zone #2: Length ~133m | Mean RSRP: -108.0 dBm | Min: -108 dBm
Diagnosis: Moderate Shadowing & Cell Edge: Signal fading near serving boundary (PCI 274).
Recommended Action: [ANTENNA_OPTIMIZATION] Sector Power Boost / Tilt Refinement (Feasibility: 75%)
Rationale: Adjusting RS (Reference Signal) power allocation or mechanical tilt may extend effective coverage radius into this edge zone.

Zone #3: Length ~390m | Mean RSRP: -107.1 dBm | Min: -108 dBm
Diagnosis: Moderate Shadowing & Cell Edge: Signal fading near serving boundary (PCI 274, PCI 25).
Recommended Action: [ANTENNA_OPTIMIZATION] Sector Power Boost / Tilt Refinement (Feasibility: 75%)
Rationale: Adjusting RS (Reference Signal) power allocation or mechanical tilt may extend effective coverage radius into this edge zone.

Zone #4: Length ~160m | Mean RSRP: -109.8 dBm | Min: -111 dBm
Diagnosis: Moderate Shadowing & Cell Edge: Signal fading near serving boundary (PCI 113, PCI 122).
Recommended Action: [ANTENNA_OPTIMIZATION] Sector Power Boost / Tilt Refinement (Feasibility: 75%)
Rationale: Adjusting RS (Reference Signal) power allocation or mechanical tilt may extend effective coverage radius into this edge zone.

Zone #5: Length ~515m | Mean RSRP: -108.1 dBm | Min: -109 dBm
Diagnosis: Moderate Shadowing & Cell Edge: Signal fading near serving boundary (PCI 362, PCI 40, PCI 24).
Recommended Action: [ANTENNA_OPTIMIZATION] Sector Power Boost / Tilt Refinement (Feasibility: 75%)
Rationale: Adjusting RS (Reference Signal) power allocation or mechanical tilt may extend effective coverage radius into this edge zone.

Zone #6: Length ~185m | Mean RSRP: -110.0 dBm | Min: -110 dBm
Diagnosis: Moderate Shadowing & Cell Edge: Signal fading near serving boundary (PCI 219).
Recommended Action: [ANTENNA_OPTIMIZATION] Sector Power Boost / Tilt Refinement (Feasibility: 75%)
Rationale: Adjusting RS (Reference Signal) power allocation or mechanical tilt may extend effective coverage radius into this edge zone.

Zone #7: Length ~538m | Mean RSRP: -110.0 dBm | Min: -110 dBm
Diagnosis: Moderate Shadowing & Cell Edge: Signal fading near serving boundary (PCI 219).
Recommended Action: [ANTENNA_OPTIMIZATION] Sector Power Boost / Tilt Refinement (Feasibility: 75%)
Rationale: Adjusting RS (Reference Signal) power allocation or mechanical tilt may extend effective coverage radius into this edge zone.

5. Explainable AI Network Anomaly Detection (Candidate Hypotheses)

Identified 76 candidate anomalies (44 Critical, 32 Warning, 0 Advisory). ML consensus: 100.0%.

- **[WARNING] Rapid Handover Ping-Pong (PCI 331 ⇄ 18)**
  Hypothesis: Candidate inadequate handover hysteresis / Time-To-Trigger (TTT) parameterization or overlapping coverage boundaries with rapid fading.

- **[WARNING] Rapid Handover Ping-Pong (PCI 331 ⇄ 4)**
  Hypothesis: Candidate inadequate handover hysteresis / Time-To-Trigger (TTT) parameterization or overlapping coverage boundaries with rapid fading.
- **[WARNING] Rapid Handover Ping-Pong (PCI 331 ⇄ 456)**
  Hypothesis: Candidate inadequate handover hysteresis / Time-To-Trigger (TTT) parameterization or overlapping coverage boundaries with rapid fading.
- **[CRITICAL] Abrupt Signal Drop (-64 dB in 4s)**
  Hypothesis: Candidate sharp shadow fading from overhead flyover, tunnel entrance, reinforced concrete structure, or antenna feeder defect.

6. Co-Channel, Adjacent-Channel & PCI Reuse Analysis

Evaluated 337 co-channel allocations, 0 adjacent carrier allocations, and 133 CRS modulo-3 collisions. Correlation: Identified 149 potential co-channel/CRS subcarrier contention pair(s). SINR metric not exposed by device modem; verification requires field scanner or network trace.

- **[Observed Same-Channel Cells] Ch 1399 | PCI 4 & 4 (Sep: 2080m)**
  Planned PCI 4 reuse on channel 1399 with healthy spatial separation of 2.1 km.
- **[Potential PCI Reuse Concern] Ch 1399 | PCI 42 & 42 (Sep: 278m)**
  Identical PCI 42 reused on channel 1399 with spatial separation of 278m (<2000m). Potential uncoordinated PCI collision hazard.
- **[Potential PCI Reuse Concern] Ch 1399 | PCI 25 & 25 (Sep: 1125m)**
  Identical PCI 25 reused on channel 1399 with spatial separation of 1125m (<2000m). Potential uncoordinated PCI collision hazard.

Interference Limitation: Note: Radio interference analysis correlates upper-layer Android TelephonyManager cell observations (carrier frequency and nominal bandwidth). Standard modems do not expose raw ACLR, spectral leakage, or RF emissions masks. Classifications indicate allocation geometry and co-channel proximity hypotheses rather than direct baseband spectrum measurements.

7. Multi-Generation Network Coverage, Continuity & Service Hole Intelligence

5G NR: 0 (0.0%) | 4G LTE: 2131 (100.0%) | 3G: 0 (0.0%) | 2G: 0 (0.0%) | OOS: 0 (0.0%)

Continuity Events: 0 total shifts (0 downgrades, 0 service losses). Generation transitions are recorded with engineering evidence without assuming unconfirmed causes.

Service Holes: 1 events identified across 7 distinct outage categories (CELLULAR_DATA_DISCONNECTED: 1).

- **[CELLULAR_DATA_DISCONNECTED] Severity: HIGH at 10:31:15.063 (BST) (58s duration)**
  Radio in service but cellular data packet bearer reported disconnected for 58.4 seconds.
- **Voice Call Reliability & Drop Risk Score: LOW (Risk Score: 0/100)**
  Evaluation: 0 call events evaluated (0 suspected drops). Explanation: Adequate cellular coverage stability. Call-drop risk indicators remain low along the tested route.

Multi-Gen Measurement Note: Voice and data bearers are independently evaluated. Telemetry gaps and test timeouts are strictly distinguished from true radio outages.

8. Methodology & Device Limitations Note:
Measurements captured via Android TelephonyManager and LocationManager. Telephony polling frequency, neighbor cell reporting, and SINR availability are device-, baseband- and carrier-dependent. Coverage surfaces and tower positions are mathematical inferences.